

THE BIOSPHERE – SUMMARY (NUMBERED VERSION)

The biosphere is the part of the Earth where all living organisms exist. It includes plants, animals, and microorganisms together with the environments that support them. The biosphere depends strongly on the soil because soil provides the essential requirements for life on land.

1. Soil as a Support for Life

- Soil supplies nutrients such as nitrogen, phosphorus, and potassium that are needed for plant growth.
- It stores water in its pores for plant use.
- It contains air spaces that provide oxygen for plant roots and soil organisms.
- It gives physical support by allowing plants to anchor firmly in the ground.

2. Moisture and Aeration

- Well-balanced soil contains both water and air in the right proportions.
- If soil contains too much water, it becomes poorly aerated and roots may suffocate.
- If soil contains too little water, plants suffer from drought stress.
- Soil therefore controls the water supply and oxygen supply to living organisms.

3. Influence on the Distribution of Plants and Animals

- Soil fertility affects the type of vegetation in an area.
- Soil texture and drainage determine how well plants can grow.
- Soil temperature also influences which species can survive.
- Since animals depend on plants, soil indirectly determines the distribution of animal life as well.

4. Role of the Biosphere in Soil Formation

- Plants add organic matter to the soil through fallen leaves and dead material.
- Animals help mix and aerate the soil as they burrow or move through it.
- Microorganisms break down organic matter and release nutrients back into the soil.
- These biological activities improve soil structure and increase soil fertility.

5. Food Chains

- Plants, which are producers, depend on soil for nutrients and water.
- Herbivores depend on plants for food.
- Carnivores depend on herbivores.

- Decomposers depend on dead plants and animals and recycle nutrients back to the soil.
- Soil forms the foundation of all terrestrial food chains.